

# Urvashi


PhD Researcher  Computational Chemistry & Materials Modelling

Swansea, United Kingdom

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ORCID  LinkedIn  GitHub  Google Scholar  Scopus



PhD researcher in Chemical Engineering working on battery electrolytes, combining density functional theory with hands-on materials characterisation. I use Python daily to build DFT and data-analysis workflows, run calculations on a Linux HPC cluster from the terminal, and turn raw instrument data (XRD, EIS) into reproducible results. Background in quantum simulation with Qiskit and Fortran; gold-medallist MSc Physics.


## Education

**PhD in Chemical Engineering (Energy Innovation)** — **Oct 2025 – Present**

*Swansea University, United Kingdom*

**Thesis title:** Development of Advanced Electrolyte Additives for Batteries.

**MSc Physics** — CGPA: 9.96/10 — **2022 – 2024**

*University of Allahabad, India* — Gold Medallist

Specialisation: Laser and Spectroscopy. Transcript  Practical report

**BSc in Mathematics, Physics and Chemistry** — 90.59% — **2019 – 2022**

*University of Allahabad, India*

Transcript

## Research Experience

**PhD Researcher** — **Oct 2025 – Present**

*Department of Chemical Engineering, Swansea University, United Kingdom* — **Supervisor:** Dr Anji Munnangi

- Running DFT calculations in Quantum ESPRESSO on the Sunbird HPC cluster to study electronic structure, ionic transport and mechanical properties of sodium-ion conducting solid electrolytes ($Na_5MSi_4O_{12}$ family).
- Combining simulation with synthesis and characterisation (XRD, SEM, EIS) and cross-checking computed results against measured data.
- Building a training dataset of electrolyte materials for machine-learning (ML) models.
- Writing Python and shell scripts to set up, run and post-process calculations and to analyse diffraction and impedance data.

**Junior Research Fellow** — **Jul 2024 – Aug 2025**

*Department of Physics, Indian Institute of Technology Roorkee* — **Supervisor:** Prof. P. C. Srivastava

- Investigated ground and excited states of $^{18}$O and $^{58}$Ni using variational quantum eigensolver methods for nuclear many-body systems.
- Developed and tested workflows in Qiskit, Python and Fortran; mapped shell-model Hamiltonians to qubits using the Jordan–Wigner transformation.
- Applied unitary coupled-cluster ansätze and benchmarked quantum results against conventional shell-model predictions for convergence and accuracy.
- Completed advanced coursework in nuclear physics, numerical analysis, computational nuclear physics and quantum computing (coursework CGPA: 9.444/10). Course transcript

## Software Projects & Outreach

- **Lab Tools** (Python, Streamlit; 2026) – built and deployed a web app for my lab: a stoichiometry calculator for synthesis precursors and an XRD plotting tool that reads Bruker `.raw` files.
- **HPC for Beginners** (2026) – writing and recording a video lecture series on using an HPC cluster (SSH, Linux commands, vi, file transfer) for the Attack on Scholar YouTube channel, which I co-founded.

## Publications

1. A. Michalak, **Urvashi**, M. A. Reddy. "Melt Quenching Process for the Synthesis of $Na_5MSi_4O_{12}$-Type Compounds, Sodium-Ion Conducting Solid Electrolytes." *ACS Omega* (2026). 10.1021/acsomega.6c08181
2. A. Michalak, **Urvashi**, T. O. Dunlop, O. Clemens, A. R. Munnangi. "Synthesis, structure, stability, and ionic conductivity of $Na_5MSi_4O_{12}$ type solid electrolytes for solid-state sodium batteries." *EES Batteries* (2026). 10.1039/D6EB00059B
3. S. Paswan, N. Jaiswal, **Urvashi**, A. Nath, M. Kumar, D. Mishra, N. Yadav. "Emerging trends and structure-activity insights of Schiff base-lanthanide complexes as antibacterial agents." *RSC Advances* **16** (2026), 3478–3508. 10.1039/D5RA08181E

## Technical Expertise

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| **Computational methods** | Density functional theory, electronic structure, phonons, CI-NEB, elastic and mechanical properties, quantum computing (VQE, UCC), machine learning |
| **Programming** | Python (NumPy, pandas, Matplotlib, Streamlit), Qiskit, Fortran, C++, Bash/shell scripting, Java |
| **HPC & environment** | Linux command line, vi, SSH/SFTP, job submission on Sunbird HPC (Supercomputing Wales) and PARAM Ganga (IIT Roorkee), Git and GitHub |
| **Scientific software** | Quantum ESPRESSO, ORCA, NuShellX, DearEIS, Gnuplot, Origin, GaussView, Avogadro, SRIM, LaTeX |
| **Experimental methods** | XRD, XPS, Raman spectroscopy, SEM, Electrochemical impedance spectroscopy, FTIR, Nanoindentation |
| **Productivity** | Microsoft Excel, Word and PowerPoint |

## Awards & Achievements

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| 2024 | **Ward Vidyant Memorial Gold Medal**, University of Allahabad, for outstanding academic performance. |
| 2024 | **Ranjana and Hari Prakash Gold Medal**, for securing the highest marks in MSc Physics (2022 – 2024). |
| 2024 | **First in order of merit, MSc Physics**, University of Allahabad. Merit certificate |
| 2024 | MHRD fellowship recipient for PhD study as a GATE-qualified candidate. GATE scorecard |
| 2022 | All India Rank 1, MSc Physics entrance examination, University of Allahabad. |
| 2022 | IIT JAM qualified. |

## Conferences & Professional Development

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| 2025 | DTP-TALENT Summer School: *Quantum Computing for Nuclear Physics*, ECT*, Trento, Italy (16 June – 4 July). Event page |
| 2025 | Workshop on Quantum Technology, IIT Roorkee. |
| 2024 | 68th DAE Symposium on Nuclear Physics, IIT Roorkee. |
| 2023 | Meghnad Saha Memorial International Conference on Frontiers of Physics, University of Allahabad. |

## Languages

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| **Medium of instruction** | English, for BSc and MSc degrees. Certificate |
| **Languages** | English (professional working proficiency); Hindi (native); Welsh (beginner) |

*References available on request.*